Unit 1 Practice Test - Version 4

90 minutes | 60 marks | Circle one answer for Questions 1-10. Show reasoning for Questions 11-18. Label all graphs. Calculator permitted; formal differentiation rules are not required.

Part A - Multiple choice (10 marks)

1. Which relation is not a function? [1]
   A. {(1,2),(2,2),(3,4)} | B. {(-1,0),(0,1),(1,0)}
   C. {(2,3),(2,5),(4,7)} | D. {(0,0),(1,1),(2,4)}

2. What is the domain of f(x) = sqrt(7 - x)? [1]
   A. x >= 7 | B. x <= 7 | C. x > 7 | D. all real numbers

3. Which interval represents -2 <= x < 5? [1]
   A. (-2,5) | B. [-2,5] | C. [-2,5) | D. (-2,5]

4. What is the vertex of y = |x + 3| - 2? [1]
   A. (3,-2) | B. (-3,2) | C. (3,2) | D. (-3,-2)

5. Which function is even? [1]
   A. x^3 + x | B. x^4 + 2 | C. x^2 + x | D. x^3 + 2

6. Relative to y=f(x), what is the horizontal change in y=f(2(x-1))? [1]
   A. stretch by 2, right 1 | B. compress by 1/2, right 1
   C. compress by 1/2, left 1 | D. stretch by 2, left 1

7. If f(x)=3x+4, what is f^(-1)(x)? [1]
   A. 3x-4 | B. 1/(3x+4) | C. (x-4)/3 | D. (x+4)/3

8. If p(x)=x+1 for x<2 and p(x)=5-x for x>=2, what is p(2)? [1]
   A. 1 | B. 2 | C. 3 | D. 5

9. What is the average rate of change of f(x)=x^2 on [1,3]? [1]
   A. 2 | B. 3 | C. 4 | D. 8

10. The instantaneous rate at a point is the slope of a: [1]
    A. horizontal axis | B. secant line | C. vertical line | D. tangent line

Optional rough work

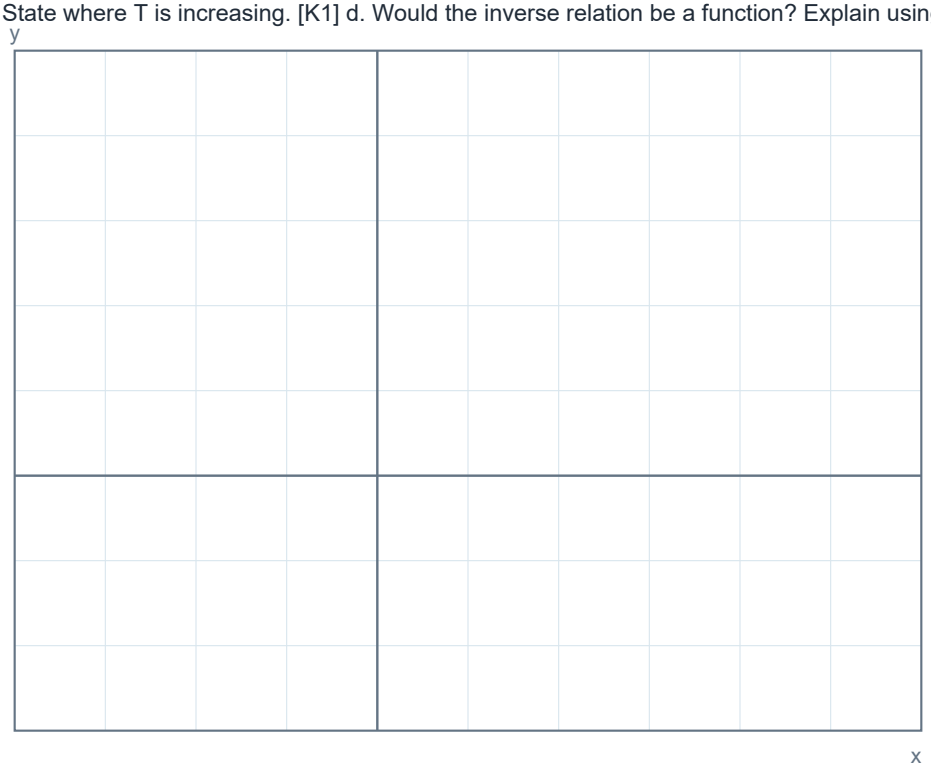
Part B - Multipart questions (50 marks)

11. Piecewise functions [9 marks]

T(x) = x + 4 if -4 <= x < 0
4 - 2x if 0 <= x <= 2
x - 4 if 2 < x <= 5

a. Sketch T. Label open and closed endpoints. [K2 C2] b. State its domain and range in interval notation. [K2]

c. State where T is increasing. [K1] d. Would the inverse relation be a function? Explain using the graph. [T1 C1]



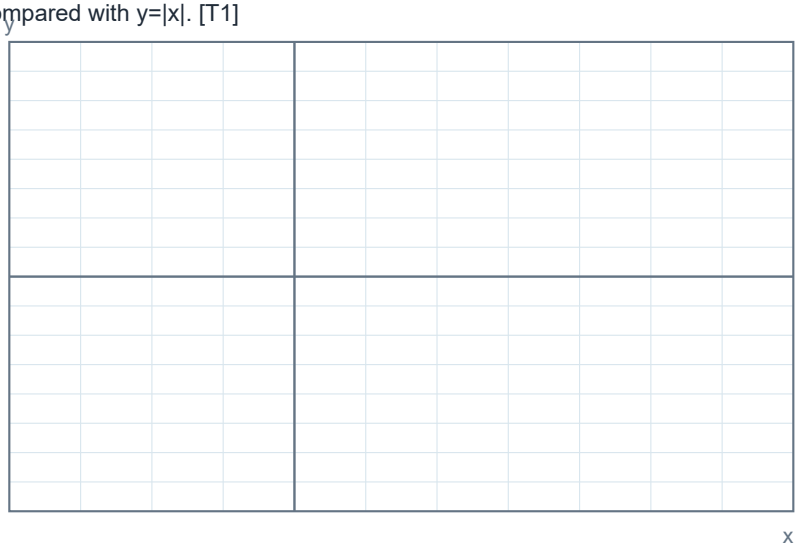
b. Domain: _____ Range: _____

c. Increasing intervals: _____

d. Inverse explanation:

12. Absolute value graph [6 marks]

Sketch g(x) = -2|x - 1| + 6. Label the vertex and both x-intercepts. [K2 C1] State domain and range. [K2] Explain the effect of -2 compared with y=|x|. [T1]



Domain/range and explanation:

13. Transforming a function [5 marks]

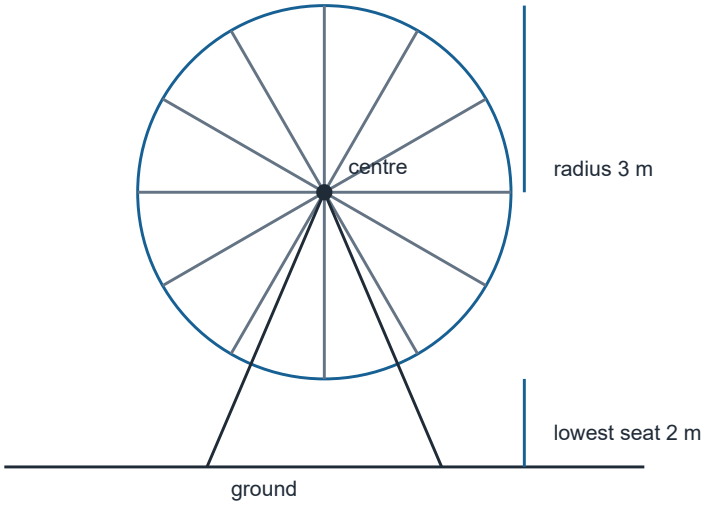
Point (16,4) lies on f(x)=sqrt(x). Let h(x)=2f(-4(x-3))+1. Find the corresponding point (show input equation) [T2]; state domain and range [K2]; describe the horizontal transformation [C1].

14. Inverse relations [5 marks]

Let q(x)=(x+1)^2+3, initially for all real x. State the inverse relation's domain and range [K2]; explain why it is not a function [T1]; restrict q's domain and write an inverse function [A2].

15. Circular ride diagram [7 marks]

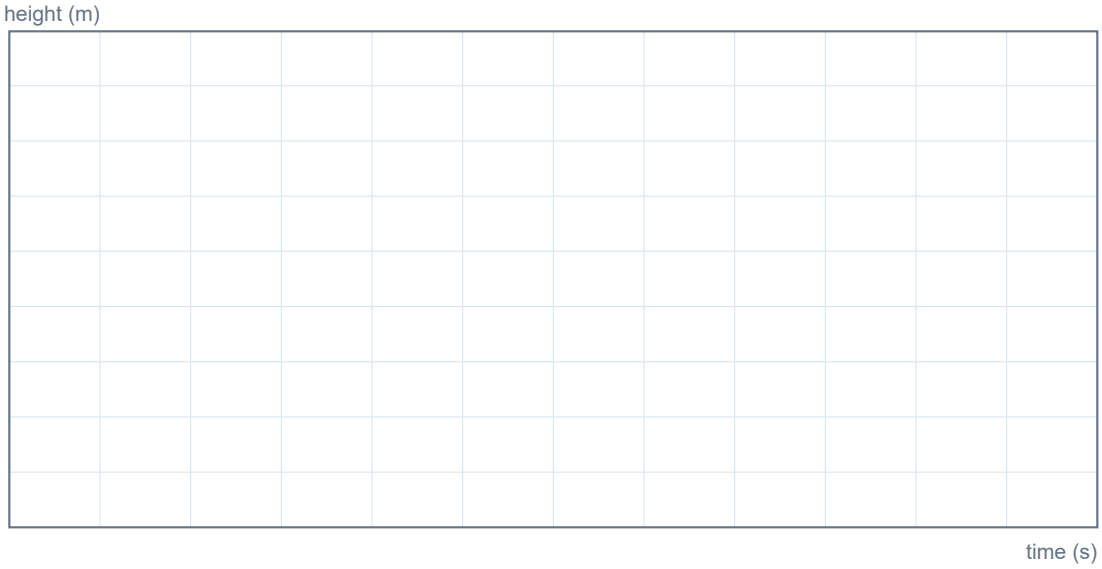
A circular ride has radius 3 m. Its lowest seat is 2 m above the ground. It makes exactly one rotation in 360 s at constant speed. A rider begins at the lowest point at t=0.



a. Find centre height and greatest seat height. [K2] b. State the height function's domain and range in interval notation. [K2]

c. A student says the greatest height is 5 m because 2+3=5. Explain the error. [T1]

d. Sketch height versus time for one rotation. Label the heights at t=0, 180, and 360 s. [C2]



Calculations and explanation:

16. Average and instantaneous rates [8 marks]

During its first 8 seconds, a ball's height is H(t)=-2t^2+16t+3 metres.

a. Calculate average rate over [0,2]. [A2] b. Estimate instantaneous rate at t=2 using [1.9,2.1]. Show both heights. [A2 T1]

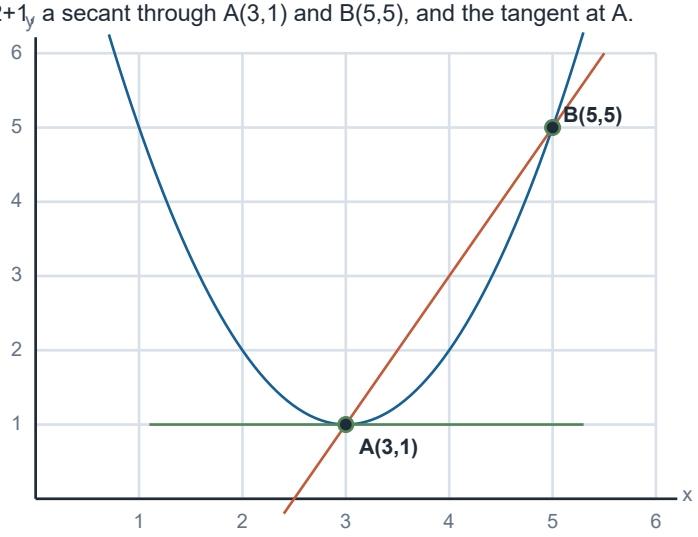
c. Find the time and height of the maximum. [K2] d. Explain why the rates in a and b differ. [C1]

17. Graph properties and interval notation [5 marks]

Let r(x)=x^5-2x. a. Show algebraically whether r is even, odd, or neither. [T2] b. Given point (2,28), name one more point using symmetry. [K1] c. Solve |x-3|<=2 in interval notation. [K2]

18. Secant and tangent diagram [5 marks]

The graph shows y=(x-3)^2+1, a secant through A(3,1) and B(5,5), and the tangent at A.



a. Which line represents average rate on [3,5]? Calculate its slope. [K1 A1]

b. Which line represents instantaneous rate at x=3? State its slope. [K1 A1]

c. Explain why the two rates differ even though both lines touch the graph at A. [T1]